

Brendan Hughes

Senior Data Scientist | Azure Data Factory & Databricks | Synapse Analytics

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SUMMARY

Senior Data Scientist with 9+ years of experience building secure, scalable data platforms, analytics pipelines, and production ML systems across GIS software, healthcare, and enterprise AI platforms. Senior Data Scientist focused on lakehouse architecture with Azure Databricks, pipeline orchestration with Azure Data Factory, and dimensional modeling with Synapse Analytics and Azure SQL, with a track record of improving reliability, backfill safety, and reporting performance under real delivery constraints. At Esri, supported geospatial analytics workloads; at Johnson & Johnson, worked in healthcare and regulated data environments; at DataRobot, delivered for enterprise AI and SaaS analytics use cases. Led engineers and cross-functional teams on migration, integration, CI/CD, testing, and security hardening efforts, with signature wins including cutting pipeline failures by roughly a third, speeding dashboard and query performance materially, and standardizing data quality controls for client-facing and executive reporting.

WORK EXPERIENCE

Senior Data Scientist | Feb 2025 – Nov 2025

Esri | Remote, Redlands, CA

- Led a small cross-functional team of data engineers, BI developers, and cloud admins during a fast-moving delivery cycle for geospatial analytics clients, where the main problem was fragmented source feeds and inconsistent refresh timing across Azure-based reporting layers. I was responsible for shaping the ingestion design, then built metadata-driven pipelines in Azure Data Factory feeding ADLS and Azure SQL, added idempotent loads, backfill controls, and DQ checks, and left runbooks and cutover notes with the support team; this stabilized daily SLAs and reduced failed reloads by roughly a third.
- Architected a lakehouse-style pattern for high-volume location and asset data where client teams needed both flexible raw storage and governed reporting models but had a limited maintenance window and could not keep reworking downstream app logic. My part was the end-to-end design and implementation: ADLS landing, Databricks Spark transforms, curated SQL-serving layers, and dimensional modeling with star schemas in Synapse Analytics. Query response for the heaviest reporting paths improved noticeably, and downstream analysts spent much less time reconciling source mismatches.
- During one sprint with a tight stakeholder demo deadline, worked directly with business analysts to clarify KPI definitions, grain, and semantic expectations for operational dashboards. I owned the modeling side, flattened nested JSON from partner APIs, normalized it into SQL-friendly structures, and built conformed dimensions in Azure SQL and Synapse so Power BI models stayed consistent. The final dashboards loaded faster and stopped breaking when source payload shapes changed.
- Cut reprocessing pain on historical backfills by introducing partition-aware Spark jobs in Azure Databricks, plus audit columns, watermark logic, and retry-safe orchestration in Data Factory. This mattered because some source systems delivered late files and duplicates, and the team needed a way to rerun weeks of data without creating reconciliation noise.
- Worked through a tricky API ingestion effort that used OAuth 2 and returned deeply nested JSON with occasional schema drift during a client rollout. I was the person accountable for getting it production ready, so I used Python, Postman, and Data Factory REST connectors to profile the payloads, documented edge cases, and built flattening logic that landed clean relational tables for downstream BI and finance reporting.

- Built CI/CD support around SQL objects, pipeline definitions, and notebook deployment using Git-based version control and test checklists, then paired with infrastructure teammates for release cutovers. The goal was not fancy process for its own sake; it was to stop last-minute promotion issues and make client changes easier to audit under security review.
- Ran design sessions and internal knowledge-sharing workshops on dimensional modeling, SLA design, and data quality patterns inside the Microsoft stack. Mentored newer engineers on when to use raw, refined, and presentation layers, how to structure star schemas, and how to avoid brittle transformations that look fine in dev but fail during real production backfills.

Senior Machine Learning Engineer | Jan 2022 – Oct 2024

Johnson & Johnson | Remote, New Brunswick, New Jersey

- Owned the data platform layer for several regulated healthcare analytics initiatives where ML, BI, and application teams all needed trusted data from a mix of internal systems and vendor APIs. I was expected to lead architecture decisions, mentor engineers, and keep delivery on track, so I designed Azure Data Factory orchestration into ADLS, Azure Databricks, and Synapse Analytics, with security-aware zone separation, lineage notes, and repeatable deployment patterns. The result was a more reliable foundation for both reporting and supervised learning workflows, and incident volume dropped meaningfully over time.
- Built data products for patient, supply, and operational reporting during periods where source teams were still changing schema and some vendor tables could not be altered because of ownership constraints. My responsibility was to make the platform resilient anyway. I used Spark on Azure Databricks to standardize ingestion, added null and range-based DQ checks, and modeled final marts in Azure SQL using dimensional techniques tuned for Power BI. Business users got steadier refreshes and fewer broken KPI definitions.
- Reduced pipeline recovery time after upstream failures by adding idempotency keys, checkpointing, and restart-safe backfill routines. This came out of a painful stretch where duplicate file drops and partial API loads were creating long cleanup cycles, especially near monthly close.
- For one critical initiative, the challenge was joining semi-structured API payloads to relational data without slowing down downstream analytics or violating audit expectations. I led the solution, used OAuth 2 integrations, flattened JSON into normalized structures, and documented transformation rules so data stewards and auditors could trace field lineage. That made compliance conversations easier and shortened the handoff time to analytics teams.
- Worked closely with ML engineers on supervised learning pipelines, mainly gradient boosting and transformer-based document workflows, where feature engineering depended on dependable batch and near-real-time feeds. My role was not just model support; I designed the upstream ingestion and serving layers with Databricks, Spark, Python, and Azure SQL so feature freshness stayed inside agreed SLAs.
- Built and tuned complex SQL used for reconciliations, semantic KPI layers, and reporting views that served finance, operations, and product teams. A few of those workloads were blocked by one hot table and ugly join patterns during month-end loads, so I rewrote parts of the transform logic, adjusted partition strategy, and reduced the worst runtime enough that the team stopped needing off-hours manual babysitting.
- Led design reviews with client-facing and internal stakeholders, translating business questions into data contracts, source-to-target mappings, and phased delivery plans. I also handled the less glamorous parts: documentation, dependency tracking, risk calls, and training analysts on how to use the curated model instead of pulling directly from raw zones.
- Built simpler support assets too: deployment checklists, on-call playbooks, and validation queries for SRE and operations partners. Those materials mattered during cutovers because the system had limited downtime windows and several teams touching it at once.
- Mentored 4 engineers across SQL standards, Spark optimization, testing discipline, and secure data handling in Azure. Some coaching was hands-on pair work during production incidents, some was through workshops

on star schemas, data contracts, and how to design lakehouse pipelines that are modular enough to survive changing requirements.

Senior Data Scientist | Jan 2016 – Nov 2021

DataRobot | Hybrid, Boston, MA

- Joined during a long growth phase where internal analytics, customer reporting, and ML experimentation were all expanding faster than the existing data foundation. I worked with product, customer success, and platform teams on a mission-critical analytics stack and took ownership of pipeline design, SQL modeling, and reporting data readiness. Using Python, SQL, Spark, and cloud warehouse patterns later aligned to Azure SQL and Synapse-style modeling, I helped standardize ingestion and create cleaner dimensional marts for executive and customer-facing use cases.
- Built ELT-style transformation workflows for usage, model telemetry, and customer activity data, with strong emphasis on testability and repeatability. The challenge was that event payloads changed often and some sources arrived late, so I added schema validation, backfill support, and freshness monitoring. That made the reporting layer more trustworthy and cut a lot of manual data cleanup.
- Helped define KPI logic and semantic reporting layers for product adoption dashboards used by leadership and go-to-market teams. I partnered with analysts to settle grain, attribution windows, and edge-case handling, then optimized the underlying SQL so dashboards loaded faster and stayed consistent from one review meeting to the next.
- One of the more involved efforts was normalizing complex JSON from application telemetry and external service feeds into relational structures that downstream teams could actually query. I owned the transformation design, used Python and Spark to flatten nested fields, documented source quirks, and created curated fact tables with conformed dimensions. That work became the basis for several recurring reports and ad hoc product analyses.
- Supported supervised and unsupervised ML workflows by preparing reliable feature datasets, monitoring drift in key inputs, and keeping training and inference-related tables aligned with business definitions. Where ML was relevant, the work touched gradient boosting, classification pipelines, and clustering-oriented analysis, but the main value I brought was dependable data preparation and traceable transforms.
- Improved performance on heavy SQL workloads by untangling a few bad join paths and overgrown aggregation layers that were causing slow executive dashboards and occasional contention during refresh windows. Performance got materially better, not perfect, but enough that refreshes stopped being a weekly fire drill.
- Built practical documentation, source mappings, and data dictionaries instead of leaving the platform as tribal knowledge. That made onboarding easier for new analysts and engineers and reduced the number of Slack messages needed to explain the same tables over and over.
- Worked in a collaborative team structure with engineers, PMs, and analysts, sometimes leading small efforts and sometimes acting as the hands-on builder. A lot of value came from steady execution: clean SQL, dependable pipelines, solid communication, and getting complex data problems into a form the business could actually use.

TECHNICAL SKILLS

- **Data Engineering Lakehouse:** Azure Data Factory, Azure Databricks, ADLS, Synapse Analytics, Microsoft Fabric, Spark, Lakehouse Architecture, Metadata-Driven Pipelines
- **Databases Modeling:** Advanced SQL, Azure SQL Database, Relational Databases, Dimensional Modeling, Star Schema Design, Data Warehousing, Query Optimization
- **Integration Transformation:** REST API Ingestion, OAuth 2.0, JSON Flattening, JSON Normalization, ELT, ETL, Data Standardization, Schema Evolution Handling

- **Reliability, Quality Security:** Idempotency, Backfills, Data Quality Checks, SLA Design, Observability, Access Control, Secure Data Architecture, Auditability
- **Analytics BI:** Power BI, Semantic Layer Design, KPI Definitions, Dashboard Performance Tuning, Reporting Marts, Executive Reporting, Self-Service Analytics
- **Programming ML:** Python, PySpark, Pandas, Scikit-Learn, Supervised Learning, Unsupervised Learning, Feature Engineering, Model Data Preparation
- **DevOps Delivery:** Git, CI/CD, Automated Testing, Deployment Checklists, Documentation, Runbooks, Agile Delivery, Stakeholder Workshops

EDUCATION

California Institute of Technology

2012 – 2014

Master's degree, Computer Science